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Rh-Catalyzed Triple Allene Approach to Bicyclo[4.4.0]decene Derivatives and Its Application for the Stepwise Synthesis of Steroid-Like Tetracyclic Skeletons

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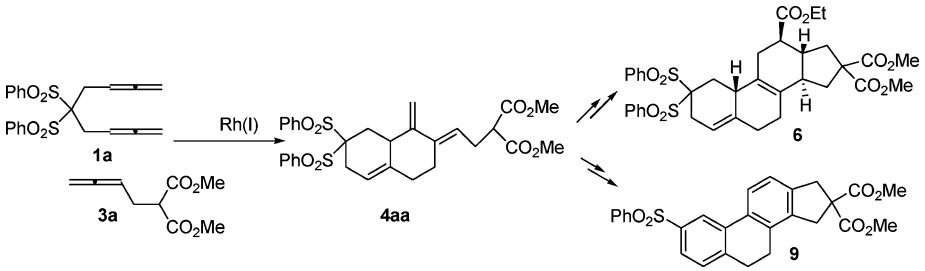
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ABSTRACT



The construction of a new type of bicyclo[4.4.0]decene products with a conjugated exocyclic diene unit was developed using Rh-catalyzed cyclization of 1,5-bisallenes in the presence of monoallenes. The steroid-like tetracyclic products could be obtained by applying an alkylation–Diels–Alder process.

Transition-metal-catalyzed cyclization reactions have shown their power for the synthesis of complex organic skeletons in an often highly selective fashion from simple and readily available starting materials.[1] Nowadays, allenes have attracted extensive attention in these reactions due to their specific synthetic versatility.[2,3] Recently, we reported the bimolecular cyclometalation of bisallenes for one-step synthesis of steroid-like tetracyclic skeletons (eq 1).[4] Our continuing study showed that the bimolecular cyclization reaction of two different bisallenes provided a mixture of four products leading to the isolation of two "cross" 18,19-norsteroid-like scaffolds formed from two different bisallenes after chromatographic separation.[5] Herein, we wish to report a Rh-catalyzed cyclization of 1,5-bisallenes in the presence of monoallenes for the efficient synthesis of fused 6,6-bicyclic

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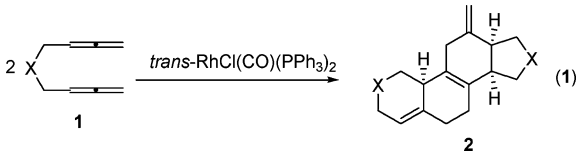
(3) For recent examples for cycloisomerizations of allenynes, see: (a) Brummond, K. M.; Painter, T. O.; Probst, D. A.; Mitasev, B. *Org. Lett.* **2007**, *9*, 347. (b) Brummond, K. M.; McCabe, J. M. *Tetrahedron* **2006**, *62*, 10541. (c) Lemière, G.; Gandon, V.; Agenet, N.; Goddard, J.; Kozak, A.; Aubert, C.; Fensterbank, L.; Malacria, M. *Angew. Chem., Int. Ed.* **2006**, *45*, 7596. For recent examples for cycloisomerizations of allenenes, see: (d) Makino, T.; Itoh, K. *J. Org. Chem.* **2004**, *69*, 395. (e) Brummond, K. M.; Chen, H.; Mitasev, B.; Casarez, A. D. *Org. Lett.* **2004**, *6*, 2161. For other examples for cyclometalations of allenes, see: (f) Wender, P. A.; Deschamps, N. M.; Sun, R. *Angew. Chem., Int. Ed.* **2006**, *45*, 3957. (g) Wender, P. A.; Croatt, M. P.; Deschamps, N. M. *Angew. Chem., Int. Ed.* **2006**, *45*, 2459. (h) Barluenga, J.; Vicente, R.; Lopez, L. A.; Tomas, M. *J. Am. Chem. Soc.* **2006**, *128*, 7050. (i) Trost, B. M.; Pinkerton, A. B.; Seidel, M. *J. Am. Chem. Soc.* **2001**, *123*, 12466. (j) Chang, H.-T.; Jayanth, T. T.; Cheng, C.-H. *J. Am. Chem. Soc.* **2007**, *129*, 4166.

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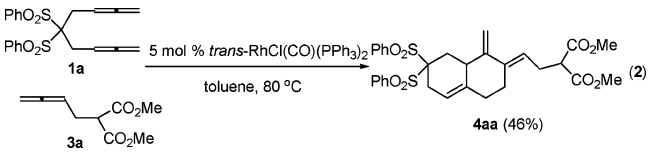
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products. In addition, with a sequential alkylation and cyclic Diels−Alder reaction, the steroid-like skeletons may be constructed in a stepwise manner diastereoselectively.



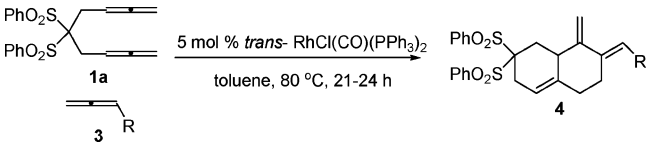
We initiated our study with *trans*-RhCl(CO)$(PPh_3)_2$-catalyzed cyclization between 1,5-bisallene **1a** and monoallene **3a** in toluene. Fortunately, after careful screening of the reaction conditions, the fused 6,6-bicyclic product **4aa** with two exocyclic carbon−carbon double bonds was isolated in 46% yield when the solution of bisallene **1a** in toluene was added via a syringe pump into the solution of rhodium catalyst and 2.5 equiv of monoallene **3a** in toluene at 80 °C (condition A) (eq 2).



Encouraged by the above result, we studied the scope of this cycloaddition reaction with different monoallenes **3** (Table 1). The reaction of **1a** and the freshly prepared phenylallene **3b** efficiently afforded the cyclization product **4ab** in 55% yield under condition B (entry 1). The structure of **4ab** was unambiguously confirmed by X-ray diffraction study (Figure 1, left).[6] The reaction of **1a** and *N*-Boc-protected buta-2,3-dienyl tosylamide **3c** provided the 2,4-dienylamide **4ac** in 55% yield (entry 2). Similarly, *N*-Boc-protected penta-3,4-dienyl tosylamide **3d** could also be used to afford **4ad** in 50% yield (entry 3). Interestingly, the reaction of the unprotected penta-3,4-dien-1-ol **3e** with **1a** afforded the alcohol **4ae** in good yield (entry 4). The reaction of buta-2,3-dienyl phthalimide **3f** or penta-3,4-dienyl phthalimide **3g** with **1a** both afforded corresponding products **4af** or **4ag** in moderate yields, respectively (entries 5 and 6).

Treatment of **4ab** with *N*-phenyl maleimide afforded the Diels−Alder product **5ab** in the yield of 91% (Scheme 1).

Table 1. Substrate Scope of Rh-Catalyzed Cyclization of Bisallene **1a** in the Presence of Monoallenes[a]



entry	R	conditions/ time (h)	yield of **4** (%)
1	Ph (**3b**)	B[b]/22	55 (**4ab**)[c]
2	$CH_2NTsBoc$ (**3c**)	A[d]/21	55 (**4ac**)[c]
3	$CH_2CH_2NTsBoc$ (**3d**)	A/24	50 (**4ad**)[c]
4	CH_2CH_2OH (**3e**)	A/22	54 (**4ae**)[e]
5	CH_2NX (**3f**)[f]	A/24	56 (**4af**)[e]
6	CH_2CH_2NX (**3g**)[f]	A/24	50 (**4ag**)[c]

[a] The reaction was carried out using **1a** (0.25 mmol), **3** (0.625 mmol), and 5 mol % of *trans*-RhCl(CO)$(PPh_3)_2$ in toluene (12 mL) at 80 °C. [b] Conditions B: a solution of **1a** and **3** in toluene (10 mL) was added to a solution of *trans*-RhCl(CO)$(PPh_3)_2$ in toluene (2 mL) via a syringe pump over ~10 h at 80 °C. [c] The byproduct **2a** was contaminated with other uncharacterized mixtures. [d] Conditions A: a solution of **1a** was added to a solution of *trans*-RhCl(CO)$(PPh_3)_2$ and **3** in toluene (2 mL) via a syringe pump over ~10 h at 80 °C. [e] The yield of byproduct **2a** was less than 18%. [f] NX = phthalimide.

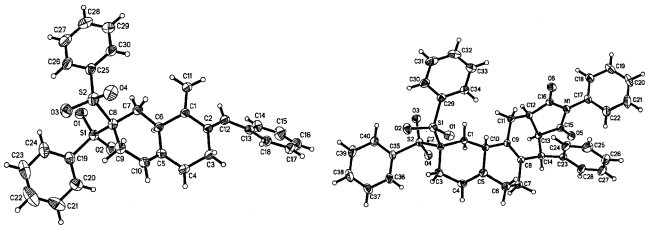
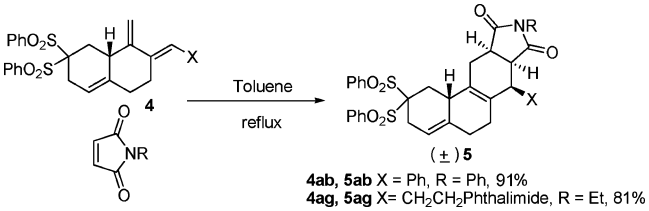


Figure 1. ORTEP representations of **4ab** (left) and **5ab** (right).

Scheme 1



The diastereoisomeric ratio should be higher than 6.3:1. The structure of tetracyclic product **5ab** was unambiguously confirmed by X-ray diffraction study (Figure 1, right).[7] Treatment of **4ag** and *N*-ethyl maleimide also afforded the tetracyclic product **5ag** in the yield of 81%. The diastereoisomeric ratio is ≥93.5:6.5.

Encouraged by this highly selective intermolecular Diels−Alder reaction, the intramolecular reaction of the 1,3-diene unit in **4aa** with an activated alkene, e.g., α,β-unsaturated

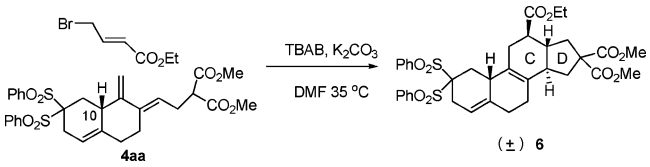
(6) Crystal data for **4ab**: $C_{30}H_{28}O_4S_2$, MW = 516.64, monoclinic, *P*2(1)/*c*. *a* = 13.4994(15) Å, *b* = 21.855(2) Å, *c* = 8.6424(10) Å, *V* = 2548.7(5) Å^3, *T* = 293(2) K, *Z* = 4. R indices (all data), R1 = 0.0844, wR2 = 0.1379. Final R indices [$I > 2\sigma(I)$], R1 = 0.0557, wR2 = 0.1269. Reflections collected/unique: 14929/5553 (R_{int} = 0.1088), number of observations [$>2\sigma(I)$] 3583; parameters, 329. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center, CCDC 651338.

(7) Crystal data for **5ab**: $C_{40}H_{35}NO_6S_2$, MW = 689.81, monoclinic, *P*2(1)/*c*. *a* = 16.4936(14) Å, *b* = 9.1500(8) Å, *c* = 23.709(2) Å, *V* = 3371.1(5) Å^3, *T* = 293(2) K, *Z* = 4. *R* indices (all data), R1 = 0.0754, wR2 = 0.1276. Final R indices [$I > 2\sigma(I)$], R1 = 0.0529, wR2 = 0.1190, Reflections collected/unique: 17938/6631 (R_{int} = 0.0940), number of observations [$>2\sigma(I)$] 4701; parameters, 442. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center, CCDC 651339.

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enoate, was also studied. It was a surprise for us to observe that when product **4aa** was treated with ethyl 4-bromocrotonate, K_2CO_3, and TBAB in DMF at 35 °C, the cyclic Diels−Alder reaction followed the allylation reaction immediately to afford the steroid-like tetracyclic product **6** in the yield of 85% (Scheme 2).[8] The diastereoisomeric ratio

Scheme 2



is ≥10:1. The structure of the major diastereoisomer, obtained easily by recrystallization from CH_2Cl_2 and petroleum ether, was unambiguously confirmed by X-ray diffraction study of **6** (Figure 2, left).[9] The C/D ring junction here

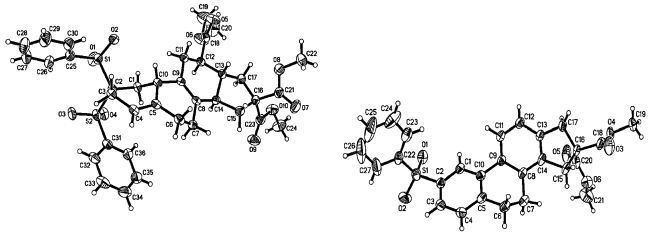
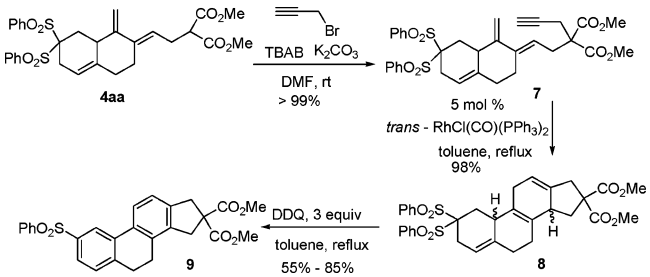


Figure 2. ORTEP representations of **6** (left) and **9** (right).

is trans, which is opposite to what was observed in our previous report.[4]

Furthermore, the intramolecular Diels−Alder reaction of the 1,3-diene unit in **4aa** and a terminal alkyne was also explored (Scheme 3). However, the reaction of **4aa**, the propargyl bromide, K_2CO_3, and TBAB in DMF afforded the alkylation product **7** in almost quantitative yield without further Diels−Alder reaction. Treament of **7** with 5 mol % of *trans*-RhCl(CO)$(PPh_3)_2$ provided the Diels−Alder product **8** as a mixture of diastereoisomeric products. Aromatization of **8** with DDQ in toluene under reflux afforded the tetracyclic product **9** with two aromatic six-membered rings in 55−85% yield,[10] indicating a desulfonylation reaction took place together with the dehydrogenation reaction.[11] The structure of aromatic product **9** was unambiguously confirmed by X-ray diffraction study (Figure 2, right).[12]

Scheme 3



In conclusion, we have developed a Rh-catalyzed cyclization of a 1,5-bisallene in the presence of a monoallene affording a new type of fused 6,6-bicyclic products with a conjugated exocyclic diene unit, which may be applied as precursors for synthesis of tetracyclic skeletons with high diastereoselectivity efficiently via the alkylation and the subsequent cyclic Diels−Alder reaction. Further study in this area is being pursued in our laboratory.

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Supporting Information Available: Experimental procedures and characterization data of all new products (PDF). This material is available free of charge via the Internet at http://pubs.acs.org.

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(8) Crystal data for **6**: $C_{36.5}H_{41}ClO_{10}S_2$, MW = 739.26, triclinic, *P*-1. *a* = 14.74(1) Å, *b* = 16.01(1) Å, *c* = 16.60(1) Å, *V* = 3550(5) Å^3, *T* = 293(2) K, *Z* = 4. R indices (all data), R1 = 0.1107, wR2 = 0.2031. Final R indices [$I > 2\sigma(I)$], R1 = 0.0688, wR2 = 0.1791. Reflections collected/unique: 18526/12890 (R_{int} = 0.0813), number of observations [$>2\sigma(I)$] 7256; parameters, 898. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center, CCDC 651340.

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(10) The ring C in compound **8** may be somewhat aromatized in the solvent if exposed to air.

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(12) Crystal data for **9**: $C_{27}H_{24}O_6S$, MW = 476.52, monoclinic, *P*2(1)/*c*. *a* = 9.3539(11) Å, *b* = 8.0181(9) Å, *c* = 31.277(4) Å, *V* = 2326.9(5) Å^3, *T* = 293(2) K, *Z* = 4. R indices (all data), R1 = 0.1305, wR2 = 0.1621. Final R indices [$I > 2\sigma(I)$], R1 = 0.0578, wR2 = 0.1198. Reflections collected/unique: 13686/9561 (R_{int} = 0.0973), number of observations [$>2\sigma(I)$] 4103; parameters, 618. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center, CCDC 651337.

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